EVENT MANAGEMENT SYSTEM

Sprint Roadmap & Project Plan

8-Week Development Plan | MVP v1 Target: End of Week 2

Student	Abdul Arfath
Reg No	P05MG24S038006
Institution	MGM College, Udupi – 576102
Internship Org	Global Zenith Tech, Udupi
Document Version	1.0	March 2026
Total Sprints	8 sprints over 8 weeks

1. Executive Summary

This document presents the complete 8-week sprint plan for building the Arfath Event Management System (EMS) — a web-based B2B event platform built with Python Flask, Firebase, PayPal, Bootstrap 5, and Docker.

The plan is divided into four phases:

- Phase 1a — MVP v1 (Weeks 1–2): Core registration and payment loop, deployable and demo-ready
- Phase 1b — MVP v2 (Weeks 3–4): Check-in, attendee portal, promo codes, and polish
- Phase 2 — Engagement (Weeks 5–6): Agenda management, speakers, analytics, and communications
- Phase 3 — Advanced (Weeks 7–8): Sponsor module, post-event automation, admin panel, and security hardening

2. Technology Stack

Layer	Technology	Purpose
Backend	Python 3.10+ / Flask	Application logic, routing, APIs
Database	Firebase Firestore	NoSQL document storage
Authentication	Firebase Auth	User identity & role management
Frontend	Bootstrap 5 + JavaScript ES6	Responsive UI
Payments	PayPal Checkout + IPN	Ticket payment processing
Email	SendGrid / SMTP	Transactional emails & QR codes
Charts	Chart.js	Organizer dashboard visualizations
Storage	Firebase Storage	Cover images, certificates, files
Deployment	Docker + Docker Compose	Containerised local-server deployment

3. Sprint Roadmap

Each sprint is two working weeks long. Priorities map directly to requirements in the SRS (v1.0).

Phase 1a — MVP v1 (Weeks 1–2)

Goal: A working, demonstrable product where attendees can register, pay, and receive a QR-coded ticket. Organizers can view registrations and basic revenue metrics.

Sprint 1 — Foundation & Authentication (Week 1)

Week 1	**Sprint 1** Foundation & Auth	• Docker + Flask project scaffolding, directory structure, .env config • Firebase Auth: sign up, login, logout, password reset (REQ-101 to 104) • Role assignment at signup: organizer / attendee / admin (REQ-103) • Venue add, edit, delete: name, address, capacity, contact (REQ-301) • Event creation: name, description, start/end dates, venue (REQ-201) • Event edit, delete drafts, and publish to public listing (REQ-202, 203) • Public event listing page — browse published events (REQ-405)

Sprint 2 — Ticketing, Payment & QR Codes (Week 2)

Week 2	**Sprint 2** Ticketing & Payment	• Ticket type creation: name, price, quantity, max per order (REQ-401) • Attendee registration flow with real-time availability validation (REQ-406, 407) • PayPal Checkout integration + IPN/webhook confirmation handling (REQ-501–504) • QR code generation on confirmed payment — unique per registration (REQ-408) • SendGrid confirmation email with QR code image attachment (REQ-409, 902) • Basic organizer dashboard: registration count, revenue total (REQ-1001) • Registration list with CSV export (REQ-1002)

🎯 **MVP v1 Deliverable (End of Week 2): Attendees can discover an event, register, pay via PayPal, and receive a QR-coded confirmation email. Organizers can view all registrations and export to CSV.**

Phase 1b — MVP v2 (Weeks 3–4)

Goal: Complete the end-to-end event lifecycle — from registration through onsite check-in — with an attendee portal, promo codes, and polished mobile responsiveness.

Sprint 3 — Onsite Check-In & Attendee Portal (Week 3)

Week 3	**Sprint 3** Check-In & Portal	• Browser-based QR scanner check-in page using device camera (REQ-801) • Scan QR code to mark attendee as checked-in with timestamp + staff ID (REQ-802) • Manual check-in: search attendee by name or email (REQ-804) • Check-in status visible on the organizer dashboard (REQ-803) • Attendee dashboard: upcoming and past events with status (REQ-601) • Attendee ticket view with QR code accessible from profile (REQ-410)

Sprint 4 — Promo Codes, Profile & Polish (Week 4)

Week 4	**Sprint 4** Promo & Polish	• Promo code creation: percentage/fixed discount, validity, usage limits (REQ-403) • Promo code validation and application during checkout (REQ-407) • User profile update: name, phone number, profile picture (REQ-105, 604) • Event cover image upload and branding color configuration (REQ-205) • Payment failure graceful handling and retry flow (REQ-506) • Bug fixes, mobile responsiveness (Bootstrap 5), and end-to-end QA (NFR-11)

🎯 **MVP v2 Deliverable (End of Week 4): Full registration-to-check-in lifecycle operational. Promo codes, attendee portal, and organizer dashboard with check-in visibility all working.**

Phase 2 — Engagement (Weeks 5–6)

Goal: Add session and agenda management, speaker profiles, analytics charts, and automated communications so the platform feels production-grade.

Sprint 5 — Agenda & Session Management (Week 5)

Week 5	**Sprint 5** Agenda & Sessions	• Session create/edit/delete: title, description, start/end time, track, location (REQ-701) • Speaker management: add/edit bio, photo, assign to sessions (REQ-702) • Virtual streaming URL per session (YouTube/Zoom embed) (REQ-703) • Public event page displays full agenda with session details (REQ-704) • Attendee session save and personal schedule management (REQ-602, 603)

Sprint 6 — Analytics & Communications (Week 6)

Week 6	**Sprint 6** Analytics & Comms	• Dashboard charts: registrations over time, revenue by ticket type (REQ-1003) • Ticket sales window: start and end date configuration per ticket type (REQ-402) • Custom registration fields: text, dropdown, checkbox with conditional logic (REQ-404) • Automated pre-event reminder emails to registered attendees (REQ-902) • Event status management: set event as cancelled or completed (REQ-204)

Phase 3 — Advanced Features (Weeks 7–8)

Goal: Deliver the differentiating Sponsor Management module, post-event automation, admin controls, and production-grade security and performance hardening.

Sprint 7 — Post-Event & Sponsor Features (Week 7)

Week 7	**Sprint 7** Post-Event & Sponsors	• Automated thank-you email with feedback survey link post-event (REQ-1201) • PDF attendance certificate generation from custom template (REQ-1202) • Sponsor deliverable tracking module: log and monitor fulfillment status • Sponsor ROI summary report generation (synopsis objective) • Event cloning: copy an existing event as a new template (REQ-206)

Sprint 8 — Admin, Reporting & Hardening (Week 8)

Week 8	**Sprint 8** Admin & Hardening	• Admin panel: view/manage all users, suspend accounts, change roles (REQ-1301, 107) • Admin global settings: configure payment gateway and email provider (REQ-1303) • Custom report generation: PDF/CSV on attendance, demographics (REQ-1004) • Webhook support for external systems (new registration, check-in events) (REQ-1102) • Security audit: Firestore rules, environment secrets in Docker, HTTPS (NFR-05 to 10) • Performance testing: validate 50+ concurrent users, Firestore index review (NFR-03, 04)

4. SRS Requirements Coverage — MVP Scope

The table below maps the Must-priority requirements from SRS v1.0 to the sprint that delivers them.

User Management & Authentication

Req ID	Requirement	Priority	MVP?
REQ-101	Sign up using email and password via Firebase Auth	**Must**	**Yes**
REQ-102	Log in and log out securely	**Must**	**Yes**
REQ-103	Role assigned at signup (organizer / attendee / admin)	**Must**	**Yes**
REQ-104	Password reset via email	**Must**	**Yes**
REQ-105	Update profile: name, phone, profile picture	Should	No

Event Management

Req ID	Requirement	Priority	MVP?
REQ-201	Create event: name, description, dates, venue	**Must**	**Yes**
REQ-202	Edit and delete draft events	**Must**	**Yes**
REQ-203	Publish event to public listing	**Must**	**Yes**
REQ-204	Set event as cancelled or completed	Should	No
REQ-205	Upload cover image and branding colors	Should	No
REQ-206	Clone an existing event as a template	Could	No

Ticketing & Registration

Req ID	Requirement	Priority	MVP?
REQ-401	Create multiple ticket types per event	**Must**	**Yes**
REQ-402	Ticket type sales start/end dates	Should	No
REQ-403	Promo codes with discount, validity, and limits	Should	No
REQ-405	Public users browse and view published events	**Must**	**Yes**
REQ-406	Attendees select ticket type, quantity, apply promo code	**Must**	**Yes**
REQ-407	Validate ticket availability and promo code eligibility	**Must**	**Yes**
REQ-408	Confirmed payment generates unique QR code	**Must**	**Yes**

REQ-409	Confirmation email with QR code sent to attendee	**Must**	**Yes**

Payment Processing

Req ID	Requirement	Priority	MVP?
REQ-501	PayPal Checkout integration for payment processing	**Must**	**Yes**
REQ-502	PayPal IPN/webhook confirmation on payment completion	**Must**	**Yes**
REQ-503	Update registration status from webhook confirmation	**Must**	**Yes**
REQ-504	Store payment records (amount, transaction ID, status) in Firestore	**Must**	**Yes**
REQ-506	Handle payment failures gracefully and allow retry	Should	No

Analytics & Reporting

Req ID	Requirement	Priority	MVP?
REQ-1001	Dashboard: total registrations, revenue, check-in count	**Must**	**Yes**
REQ-1002	View registrations list and export to CSV	**Must**	**Yes**
REQ-1003	Charts: registrations over time, sales by ticket type	Should	No
REQ-1004	Custom reports: PDF/Excel on attendance and demographics	Should	No

5. Non-Functional Requirements

ID	Requirement	Sprint
NFR-01	Public event list and registration page loads in < 3 seconds	Sprint 2
NFR-02	Registration and payment completes within 5 seconds (excl. PayPal redirect)	Sprint 2
NFR-03	Support at least 50 concurrent users in MVP; 500+ in later releases	Sprint 8
NFR-04	Firestore queries use indexes for < 200ms response on common queries	Sprint 8
NFR-05	All communication over HTTPS (SSL/TLS)	Sprint 8
NFR-06	Passwords managed by Firebase Auth; never stored in Firestore	Sprint 1
NFR-07	No card data stored in the system; handled entirely by PayPal	Sprint 2
NFR-08	Firestore security rules: organizers can only modify their own events	Sprint 8
NFR-09	Flask routes validate authentication and role-based authorization	Sprint 1
NFR-10	API keys and secrets stored in .env and Docker secrets — never hard-coded	Sprint 1
NFR-11	Responsive UI on desktop, tablet, and mobile (Bootstrap 5)	Sprint 4
NFR-15	System uptime target: 99.5% (excluding planned maintenance)	Ongoing
NFR-20	Code follows PEP 8 standards with documentation	Ongoing
NFR-21	Git version control with clear commit messages	Sprint 1

6. Risks & Mitigations

Risk	Severity	Mitigation
PayPal IPN/webhook setup complexity	**High**	Set up ngrok early in Sprint 2 to expose a local endpoint for testing. Use PayPal Sandbox throughout development.
Firebase Firestore security rules inadequate before go-live	**High**	Draft security rules in Sprint 1 alongside the data model. Audit and finalize in Sprint 8 before any external access.
QR code scanner requires HTTPS for camera access on mobile	**Medium**	Ensure Docker deployment has SSL configured (even self-signed) before Sprint 3 check-in testing.
Firebase Auth and Firestore service outages	**Medium**	Implement retry logic (REQ-NFR-17) and graceful degradation. Show user-friendly error states instead of blank screens.
PayPal payment race conditions (duplicate registrations)	**Medium**	Idempotency check on IPN webhook: validate transaction ID against Firestore before creating registration.
Scope creep delaying MVP v1	**High**	Strictly enforce MVP scope in Sprints 1–2. All 'Should' and 'Could' requirements are deferred to Sprint 3+.
Local Docker deployment lacks cloud DNS/SSL	**Low**	Use static IP or mDNS hostname on the local network. Document SSL setup for organizer self-hosting.

7. Future Scope (Post 8-Week Cycle)

The following features are planned for releases beyond the 8-week cycle, as defined in SRS v1.0 Section 6:

- Mobile application (React Native) with offline capabilities
- CRM integration: HubSpot and Salesforce sync via API (REQ-1101)
- SMS notifications via Twilio (REQ-904)
- Exhibitor management module with booth-level lead capture
- Multi-currency and multi-language support (REQ-505)
- Attendee ticket transfer (REQ-411) and waitlisting (REQ-412)
- Refund processing from the organizer dashboard (REQ-413)
- White-labeling: custom domains and full brand control
- Advanced networking: matchmaking and virtual meeting rooms
- AI-powered session recommendations and chatbot assistant
- Blockchain-based ticketing for anti-counterfeiting
- Sustainability tracking: carbon footprint estimation per event

8. References

[1] Jeny et al. (2022). A web-based college event management system and notification sender. ICIRCA, IEEE.

[2] Liew & Tan (2021). QR code-based student attendance system. ICSGRC, IEEE.

[3] Shah et al. (2023). Event management systems (EMS). Journal of Applied Technology and Innovation, 7(2).

[4] Chavan (2022). A review on Firebase for mobile application development. IJRASET, 10(1).

[5] Hadiwiyanti et al. (2022). Evaluation of campus EMS using system usability scale. ITIS, IEEE.

[6] Chen et al. (2023). Using Docker container technology for rapid deployment. ICEIB, IEEE.

[7] Ferreira et al. (2024). Accelerating cloud deployment through containerization. SMC, IEEE.